INCH-POUND

MIL-M-13331A(AR)
9 July 1990
SUPERSEDING
MIL-M-13331(ORD)
09 October 1940

MILITARY SPECIFICATION

MOUNT, TELESCOPE: M18A1

This specification is approved for use by the U.S. Army Armament, Munitions and Chemical Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a mount used to hold a panoramic telescope of the M12 Series for laying the gun in azimuth.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this specifications to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation. (See 6.1).

SPECIFICATIONS

MILITARY

MIL-F-13926	-Fire Control Materiel, Manufacture and Inspection, General Specification for
MIL-P-14232	-Parts, Equipment and Tools for Ordnance Materiel, Packaging of
MIL-I-45607	-Inspection Equipment, Acquisition, Maintenance and Disposition of

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army ARDEC, ATTN: SMCAR-BAC-S, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A
FSC 1240

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STANDARDS

MILITARY

MIL-STD-109 -Quality Assurance Terms and Definitions

(Unless otherwise indicated, copies of federal and military specifications, standards and handbooks are available from: Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issue shall be those in effect on date of the solicitation.

DRAWINGS

U.S. Army Armament Research, Development and Engineering Center (ARDEC)

7295505	Perpendicularity Gage
7578402	Mount, Telescope: M18A1
8658940	Gunner Quadrant, Calibrated

PACKAGING DATA SHEET

7578402	Packaging of Mount, Telescope: M18A1

(Copies of drawings and packaging data sheets required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede the applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified, a sample shall be subjected to first article inspection (see 4.2 and 6.1).

3.2 Materials. Materials shall be in accordance with drawings, material specifications, and general specifications forming a part of this specification.

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3.3 Fabrication. The mount shall be manufactured in accordance with Drawing 7578402 and drawings pertaining thereto.

3.4 General specifications. The contractor shall be responsible for the compliance with the requirements of specification MIL-F-13926.

3.5 Vibration. The mount shall be vibrated between 2 to 2-1/2 minutes at an amplitude of not less than 1/16 inch (1/8 inch total movement) and a frequency of 30 cycles per second after which it shall meet all requirements.

3.6 Adjustment. With the Mount, Telescope, M18A1 mounted so that the longitudinal axis of the actuating arm is horizontal, the bracket vertical as shown on assembled view drawings, and the axis of the socket for the panoramic telescope vertical, the mount shall be adjusted as follows:

a. The elevation level bubble shall be adjusted to show the bubble central with respect to the level vial graduations.

b. The line of sight adjusting screw shall be adjusted to bring the line of sight of a suitable M12 series panoramic telescope (or collimating gage and telescope) set at "0" deflection, parallel to the planes of the attaching surfaces of the bracket.

c. The cross level vial shall be adjusted to show the bubble control with respect to its graduations when the mount is set up as specified in 3.6, and shall be adjusted by means of the cross level worm to such position that when the actuating arm is rotated the deviation of the line of sight of the panoramic telescope from the point of intersection on a target is zero, or in case of an instrument with inherent error, minimum.

3.7 Accuracy. With the mount set up and cross leveled as specified in 3.6, the mount shall meet the following requirements:

3.7.1 Axis of socket. The axis of the socket for the panoramic telescope shall be vertical, in the cross level plane, within a tolerance of 1/2 mil.

3.7.2 Actuating arm deviation. The intersection of the crosslines of the reticle shall not deviate more than 3/4 mil, either horizontally or vertically, from a point on a target when the actuating arm is rotated from 15° depression to 65° elevation.

3.7.3 Vertical deviation. The intersection of the crosslines of the reticle shall not deviate from a plumb line more than 1/2 mil when the vertical travel of the telescope and mount is accomplished by means of the worm through maximum travel permitted by the rocker.

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3.7.4 Backlash. Backlash in the worm and worm wheel mechanisms shall not exceed the following limits:

a. In the rocker worm mechanism, 3 minutes elevation displacement

b. In the cross level worm mechanism, 2 minutes cross level displacement

3.7.5 Azimuth correction. With a panoramic telescope mounted in place and using a plumb line as a target, the "Azimuth Correction Test" shall show readings on the azimuth scale of the panoramic telescope as tabulated in the Azimuth Correction Column below:

TABLE I. Azimuth Correction

Angle of Cross level (Cant) E	Elevation Angle X	Azimuth Correction Y	Tolerance
Degrees Left or Right	Degrees	Degrees	
8	30	4°35'38"	±1.5 mils(5'-5")
8	45	7°55'23"	±2.0 mils(6'-45")
8	65	16°37'5"	±2.5 mils(8'-45")

NOTE. The Mount, Telescope, M18A1 automatically sets off the amount of azimuth correction necessary in gun pointing due to elevating the gun with the trunnions out of level. The correction varies with the cross level angle and the angle of elevation and is calculated by means of the following formula:

Tan Y = tan X sin E in which

Y = azimuth correction

X = elevation angle (=angle turned by actuating arm about its longitudinal axis, starting with axis of actuating arm pivot horizontal)

E = angle of cross level (=angle between axis of actuating arm and the horizontal)

3.7.6 Moving parts. All moving parts shall function smoothly and properly without irregularities, undue friction or looseness.

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3.7.7 Electrical system. The electrical system shall function properly and shall illuminate the bubbles on the mount, the azimuth scale and micrometer, and the crosslines of the reticle of the panoramic telescope, so that they are plainly visible in a dark room.

3.8 Operating temperatures. The mount shall be operable at -40° ± 5° Fahrenheit (F) (-40° ± 2° Centigrade (C)) and at +150° ± 5°F (±65° ± 2°C). Subsequent to the temperature tests the mount shall meet all of the requirements of this specification at room temperature.

3.9 Workmanship. The workmanship of MIL-F-13926 shall apply.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspection set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 General provisions. The component and subassembly inspection requirements of MIL-F-13926 form a part of the quality assurance provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 First article (initial production) approval. The requirement for first article approval and the responsibility (Government or contractor) for first article testing shall be as specified in the contract. The sample for first article approval tests shall consist of three (3) mounts plus three (3) each of all items covered by Quality Assurance Provisions (QAPs). The

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sample shall be manufactured in the same manner, using the same materials, equipment, processes, and procedures as used in regular production. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as used in regular production. The 3 mounts shall be tested in accordance with, and meet the requirements of Tables I and II.

4.2.1 Government testing. When the Government is responsible for conducting first article approval tests, the contractor, prior to submitting the sample to the Government, shall inspect the sample to insure that it conforms to all requirements of the contract and submit a record of this inspection, including certificates of conformance for materials.

4.2.2 Contractor testing. When the contractor is responsible for conducting first article approval tests, the sample shall be inspected by the contractor for all the requirements of the contract. The sample and a record of this inspection, including certificates of conformance for materials, shall be submitted to the Government for approval. The Government reserves the right to witness the contractor's inspection.

4.3 Examination and tests.

4.3.1 Components and subassemblies. All components and subassemblies shall be inspected in accordance with the inspection provisions contained in QAPs listed in the Technical Data Package (TDP). In the absence of QAPs, the applicable Quality Assurance Provisions of MIL-F-13926 shall apply.

4.3.2 Final acceptance inspection. Subsequent to first article approval, examination and tests related to Section 3 herein shall be performed. The tabulated classification of defects in Tables I and II shall constitute the minimum inspection to be performed by the supplier after first article approval and prior to government acceptance or rejection by item or lot.

4.3.3 Classification of characteristics. Quality conformance examinations and tests are specified in the following Classification of Characteristics paragraphs. The contractor's quality program or detailed inspection system shall provide assurance of compliance of all characteristics with the applicable drawing and specification requirements utilizing as a minimum the conformance criteria specified herein.

4.3.4 Alternative inspection provisions. Alternative inspection procedures, methods, or equipment, such as statistical process control, tool control, other types of sampling procedures, etc., may be used by the contractor when they provide, as a minimum, the level of quality assurance required by the provisions specified herein. Prior to applying such alternative procedures,

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methods, or equipment, the contractor shall describe them in a written proposal submitted to the procuring contracting officer for evaluation and approval by the Government. When required, the contractor shall demonstrate that the effectiveness of the proposed alternative(s) is equal to or better than the specified quality assurance provisions herein. In cases of dispute as to whether the contractor's proposed alternative(s) provide equal assurance, the provision of this specification shall apply. All approved alternative inspection provisions shall be specifically incorporated into the contractor's quality program or detailed inspection system, as applicable.

TABLE II. Requirements and test procedures.

No.	Characteristic	Requirement	Test procedure
Critical:	None		
Major:	100% Inspection		
101.	Vibration	3.5	4.6.1
102.	Adjustment	3.6	4.6.2
103.	Axis of socket	3.7.1	4.6.3
104.	Actuating arm deviation	3.7.2	4.6.4
105.	Vertical deviation	3.7.3	4.6.5
106.	Backlash	3.7.4	4.6.6
107.	Azimuth correction	3.7.5	4.6.7
108.	Moving parts	3.7.6	4.6.8
109.	Electrical system	3.7.7	4.6.9
110.	Workmanship	3.9	MIL-F-13926
111.	Packaging	5.1	4.8
Minor:	None		

4.4 Special sampling.

4.4.1 General. Subsequent to meeting the requirements of Table I, three quadrants shall be selected at random by a Government representative as a special sample from each 50 produced or from each month's production, whichever occurs first. The samples shall meet the requirements and tests in Table I after being subjected to Table II testing.

TABLE III. Requirements and Test Procedures.

NO.	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
301.	Operating temperature	3.8	4.7

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4.4.2 Failure of sample. Should any one item of a special sampling fail to meet the specified test requirements, acceptance of the represented inspection lot will be suspended by the Government until necessary corrections have been made by the contractor and the resubmitted samples have been approved.

4.5 Test equipment.

4.5.1 Inspection equipment. Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607.

4.5.2 Government furnished inspection equipment. Where the contract provides for Government furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements specified in MIL-I-45607.

4.5.3 Contractor furnished inspection equipment.

4.5.3.1 Government design. Unless otherwise specified in the contract, all inspection equipment specified by drawing number in specifications or SQAP forming a part of the contract shall be supplied by the contractor in accordance with technical data listed in the list of Inspection Documents when provided with the Technical Data Package (TDP).

4.5.3.2 Contractor design. The contractor shall design and supply inspection equipment compatible with the requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions, and tolerance used in the design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit, and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.5.3.3 Test equipment. In conjunction with 4.5.3.2, the following standard test equipment shall be utilized in the performance of the applicable test as specified in 4.7.

NOMENCLATURE	DESCRIPTION
1. Hot and cold chamber	Standard type conforming to the accuracy outlined in test facilities specified under test facilities of MIL-F-13926.

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4.6 Methods of inspection.

4.6.1 Vibration. The mount shall be positioned on a fixture as shown in the assembled view of the drawing and vibrated in accordance with the frequency, amplitude and time of 3.5. At the conclusion of the test the mount shall be subjected to a visual and tactile examination, and shall meet the requirements of 3.6 through 3.9.

4.6.2 Adjustment. The mount shall be positioned on the fixture as described in 3.6. The elevation and cross level vial bubbles shall be centered and the line of sight shall be as described in 3.6b, to meet the requirements in 3.6.

4.6.3 Axis of socket. The axis of the socket shall be tested by utilizing Perpendicularity Gage, 7295505, except that find number 1, Body, shall be 5 1/2 inches in diameter and the top surface shall be held flat and parallel to within .0002 of the 9/32 dimension surface and the top surface to be perpendicular to the .7495 diameter, Datum "A" to within .0002. (The "V" does not have to be machined into find 1.) The gage shall be inserted in the socket and calibrated gunners quadrant, 8658940, shall be used to verify that the axis of the socket meets the requirement of 3.7.1.

4.6.4 Actuating arm deviation. The actuating arm shall be rotated to the depression stop and brought back to 15° in depression. The reticle of panoramic telescope or collimating gage and telescope shall be in coincidence with the target. Rotate the actuating arm to 65° elevation or horizontally more than 3/4 mil to meet the requirements of 3.7.2.

4.6.5 Vertical deviation. Turn the rocker until the stop is reached. The reticle of the telescope shall be placed in coincidence with the plumb line and turn the rocker knob until the other stop is reached and assure that the reticle does not deviate from the plumb line more than 1/2 mil to meet the requirements of 3.7.3.

4.6.6 Backlash. Place the modified gage, 7295505, in the socket of the mount. Turn the worm knobs in the right hand direction to center level vial bubbles. Take a reading in the pitch direction with the gunners quadrant, 8658940. Scribe an index line on the housing of the rocker and line a pointer up with the index line secured to the knob. Turn knob in the right hand direction one full turn, reverse the knob and line up the pointer with the index line, and take another reading with the gunners quadrant. The difference between the first and second reading is the backlash, and must not exceed the requirements of 3.7.4a. Follow the above procedure for the cross level mechanism, and the backlash must not exceed the requirement of 3.7.4b.

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4.6.7 Azimuth correction test. The mount, with a panoramic telescope in place in its socket shall be attached to a surface which is in a position to hold the axis of the actuating arm at the angle of cross level (E) with the horizontal. The mount shall be leveled by means of the worms and level bubbles on the mount. The axis of the actuating arm pivot shall be brought into a horizontal position. The panoramic telescope shall be directed at a plumb line target. The actuating arm shall be rotated through the angle equal to the elevation angle (X), measured in a plane perpendicular to the axis of the actuating arm. The azimuth correction angle (Y) through which the head of the panoramic telescope must be turned to bring the line of sight back into coincidence with a plumb line shall be measured on the azimuth circle of the panoramic telescope, and shall not exceed the tolerances of 3.7.5.

4.6.8 Moving parts. All moving parts shall meet the requirements of 3.7.6.

4.6.9 Electrical system. The electrical system shall meet the requirements of 3.7.7.

4.7 Operating temperatures. All moving parts shall be operational and functional at the extreme temperatures of 3.8.

4.8 Packaging inspection. The preservation, packing, and marking shall be inspected to verify conformance to the requirements of 5.1.

5. PACKAGING

5.1 Packaging, packing and marking. Packaging, packing, and marking shall be in accordance with MIL-P-14232 and Packaging Data Sheet 7578402. The level of protection shall be as specified in the procurement document.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

6.1 Acquisition requirements. Procurement data should specify the following:

a. Title, number and date of this specification.

b. Selection of an applicable level of preservation, packaging, and packing (See 5.1).

c. Applicable packaging data sheet number (See 5.1).

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d. Applicable stock number.

e. Provisions for First Article Testing.

f. Contract data requirements for submission of inspection equipment designs conforming to Data Item Description DI-R-1714. (See 6.2.1.)

6.2 Inspection equipment design.

6.2.1 Submission of designs for approval. Contractor designs for final acceptance inspection shall be approved by the Government prior to fabrication or procuring the equipment. The contractor is referred to MIL-HDBK-204 for guidance. Submission of design concept on inspection equipment is permissible for tentative approval. The completion date for design review will be based on the data of the final submission of designs and the required delivery schedule as stipulated in the contract. Submit designs as required to: Commander, U.S. Army Armament Research, Development and Engineering Center, ATTN: AMSMC-QAF-I(D), Picatinny Arsenal, NJ 07806-5000. This address will be specified on the Contract Data Requirements List DD Form 1423 in the contract. Unless otherwise specified, data item DI-R-1714 will apply. When the contractor submits inspection equipment designs to the Government for approval, he shall give the following information in his letter of transmittal:

a. The contract number.

b. The contract item (name, model number, etc.).

c. The designs remaining to be submitted and the expected date of submittal.

6.3 Drawings. Drawings listed in Section 2 of this specification under the heading US Army Armament, Research Development and Engineering Center (ARDEC) may also include drawings prepared by, and identified as Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal, or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under the cognizance of ARDEC.

6.4 Subject term (key word) listing.

Fire control
Mount

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6.5 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army-AR

Preparing Activity:
Army-AR

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INSTRUCTIONS

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2. The submitter of this form must complete blocks 4, 5, 6, and 7.
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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-M-13331A (AR)	9 July 1990

3. DOCUMENT TITLE

MOUNT, TELESCOPE: M18A1

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial	(2) AUTOVON
US Army ARDEC Standardization & Specification Office	(201) 724-6674	880-6674
c. ADDRESS (Include Zip Code) ATTN: SMCAR-BAC-S/Bldg 6 Picatinny Arsenal, NJ 07806-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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